

ANALYSTLAB AFRICA

Machine Learning Internship Programme

Machine Learning Proposal

1. Problem Type

Binary classification. The target variable, Churn, takes two values (Yes / No), so the task is to predict class membership, not a continuous quantity — ruling out regression approaches.

2. Target and Features

Target: Churn (binary).

Features (20, after dropping the customerID identifier): a mix of binary categorical (gender, Partner, Dependents, PhoneService, PaperlessBilling), multi-class categorical (MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaymentMethod), and numeric (tenure, MonthlyCharges, TotalCharges, SeniorCitizen) fields.

3. Recommended Algorithms

Algorithm	Why it fits	Trade-off
Logistic Regression	Fast, interpretable baseline; coefficients directly show which factors raise or lower churn risk — useful for explaining predictions to the retention team.	Assumes roughly linear relationships; may underfit complex interactions between features.
Random Forest	Handles mixed categorical/numeric data and non-linear interactions well without heavy preprocessing; robust to outliers; gives feature importance.	Less interpretable than logistic regression; can be slower to tune.
Gradient Boosting (XGBoost / LightGBM)	Typically the strongest performer on tabular churn-style data; handles imbalance well with built-in weighting.	More hyperparameters to tune; higher risk of overfitting without careful validation.
K-Nearest Neighbours / SVM (secondary)	Useful as additional comparison points, especially after feature scaling.	Scale-sensitive and slower on larger feature sets; usually outperformed by tree-based models here.

Recommended approach: start with Logistic Regression as an interpretable baseline, then benchmark Random Forest and a Gradient Boosting model against it. Select the final model based on the evaluation metrics in Section 4, not accuracy alone.

4. Evaluation Metrics

Because the classes are imbalanced (~73.5% No vs. ~26.5% Yes), accuracy alone would reward a model that simply predicts "No" for everyone. The following metrics are recommended instead:

- Precision — of customers flagged as likely to churn, how many actually churn (controls wasted retention spend).
- Recall — of customers who actually churn, how many the model catches (controls missed at-risk customers).
- F1-score — harmonic mean of precision and recall, useful for a single comparison number.
- ROC-AUC — threshold-independent measure of how well the model ranks churners above non-churners.

The retention team will ultimately choose an operating threshold based on the cost of a false positive (an unnecessary retention offer) versus a false negative (a churned customer who was never flagged).

5. Preprocessing Strategy

- Clean TotalCharges: coerce to numeric and impute the 11 missing values (all tenure = 0 customers) as 0.
- Drop customerID (identifier only, no predictive signal).
- Encode categorical variables: binary-encode two-level fields (e.g. gender, Partner); one-hot encode multi-level fields (e.g. Contract, PaymentMethod, InternetService).
- Scale numeric features (tenure, MonthlyCharges, TotalCharges) — particularly important for KNN/SVM and beneficial for regularised Logistic Regression.
- Stratified train/test split (e.g. 80/20) to preserve the ~73.5% / 26.5% class ratio in both sets.
- Address class imbalance directly — via `class_weight='balanced'` in scikit-learn models and/or SMOTE oversampling on the training set only (never on the test set, to avoid leakage).
- Use k-fold cross-validation (stratified) during model selection and hyperparameter tuning.

6. Next Steps (Coming Weeks)

- Week 2: Implement the preprocessing pipeline and train baseline Logistic Regression, Random Forest, and Gradient Boosting models.
- Week 3: Hyperparameter tuning, cross-validation, and metric comparison across models.
- Week 4: Model interpretation (feature importance / SHAP), business recommendations, and final reporting to the client.